

V Sree Raghu Vardhan

SOFTWARE ENGINEER

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SUMMARY

Software Engineer with experience building production AI backend systems using Python, FastAPI, LangChain, Azure OpenAI, PostgreSQL, MongoDB, Redis, and REST APIs. Built enterprise AI agents, RAG pipelines, semantic retrieval systems, structured output validation, and scalable backend services for HRMS workflows. Experienced in designing reliable APIs, integrating enterprise systems, and shipping production-grade AI features with CI/CD and open-source contributions.

EXPERIENCE

Darwinbox – Hyderabad, India

07/2025 – 04/2026

SDE-1

- Developed production-grade backend services and REST APIs using **Python, FastAPI, MongoDB, Redis**, and **SQL** to power AI-driven enterprise HRMS workflows.
- Built **LangChain-based multi-step AI agent workflows** integrating enterprise APIs, MCP tools, semantic retrieval, document processing, and business automation across multiple HR modules.
- Designed structured output validation layers using **Redis-backed state management** to preserve original API responses across agent steps, reducing LLM hallucinations during complex form completion workflows.
- Designed reusable backend components for complex dynamic enterprise forms supporting hundreds of configurable fields across multiple HRMS modules, simplifying future feature development.
- Built semantic retrieval pipelines using embeddings to identify relevant profile fields before tool execution, improving response accuracy while significantly reducing LLM context size and token usage.
- Integrated **Azure OpenAI** and **OpenAI** APIs for document understanding, image parsing, and enterprise workflow automation, enabling AI-assisted form completion and document processing.
- Implemented resilient agent workflows with asynchronous streaming, retry handling, error recovery, and enterprise API integrations to improve reliability and user experience.
- Implemented asynchronous streaming by merging multiple concurrent LLM response streams into a unified real-time client experience using LangChain callback handlers and asynchronous task orchestration.
- Built **LangSmith** evaluation pipelines with reproducible testing infrastructure using recorded API responses, database snapshots, mocked dependencies, and deterministic datasets to evaluate agent behavior consistently.
- Extended the platform's Retrieval-Augmented Generation (RAG) framework by refactoring source ingestion pipelines for extensibility, enabling support for future knowledge source types.
- Optimized backend performance by reducing MongoDB queries through intelligent prefetching and request-level caching, improving latency for high-traffic agent workflows.
- Established the platform's automated testing framework by introducing unit testing, reusable fixtures, mocked dependencies, and GitLab CI coverage reporting to improve software quality and deployment confidence.
- Collaborated closely with frontend engineers, product managers, and business analysts to design scalable AI features, troubleshoot production issues, and deliver enterprise-grade backend solutions.

Darwinbox – Hyderabad, India

01/2025 - 06/2025

SDE Intern

- Built a production Business Requirements Document generation platform using FastAPI, Azure/OpenAI LLM APIs, structured document templates, and automated document generation workflows.
- Developed an internal Jira-based performance analytics solution to track workflow metrics, engineering productivity, and operational reporting.
- Researched and prototyped enterprise document management solutions, including browser-based DOCX rendering and editing capabilities.

Providence – Hyderabad, India

06/2024 - 08/2024

Summer SDE Intern

- Built ETL pipelines processing Azure log data, transforming operational telemetry into structured datasets for reliability analysis and production issue identification.
- Worked with Django, Node.js, Azure PowerShell and React

OPEN SOURCE

Python Packaging Authority (PyPA) — Bandersnatch

- Authored 11+ merged pull requests accepted by PyPA maintainers across performance optimization, architecture refactoring, feature development, bug fixes, and test infrastructure.
- Refactored core mirroring and verification workflows, simplifying the codebase and improving maintainability.
- Identified major optimizations in hash calculation logic and S3 implementations which reduced mirror times by 41%.
- Modernized and expanded the test suite by refactoring legacy unit tests and improving test organization.
- Collaborated with PyPA maintainers through design discussions, code reviews, production bug fixes, and comprehensive test improvements.

EDUCATION

B.E Information Technology - 05/2025

Chaitanya Bharati Institute of Technology, Hyderabad, India

- CGPA – **9.28**
- Student Lead – Algorand Blockchain Club
- 1st Prize – Paper Presentation on Simultaneous Ride Availability Analysis

Intermediate, MPC – 05/2021

FIITJEE Jr. College, Hyderabad, India

- JEE Mains - **97.36 percentile**
- JEE Advanced – **11515**
- EAMCET - **1946**

SKILLS

Languages

Python • Java • SQL • JavaScript

Backend

FastAPI • REST APIs • Pydantic • Async Python

Databases

MongoDB • Redis • PostgreSQL

AI Engineering

LangChain, LangSmith, Azure OpenAI, OpenAI API, MCP, RAG, Semantic Search, Pinecone, LangGraph, AI Agents, Vector Search

Cloud & DevOps

AWS (ECS, S3), Docker • Git • GitHub • GitLab CI • Jenkins • Kubernetes • Prometheus • ElasticSearch

Core

Data Structures & Algorithms • Distributed Systems • System Design • Unit Testing • Debugging